Dynamic-Payload Pick-and-Pour Control for a UR5 Barista Robot: Progress Report

Soumaya El Mansouri, Jingxuan Yang, Tracy Yang
RBE 501 Robot Dynamics

Problem

- Pouring is hard for robot arms.
- The held pitcher loses mass as it empties, and the liquid shifts.
- A poor grasp or poor sensing of the poured amount causes spills or overfilled cups.
- Our focus is the changing payload and the sensing problem, in simulation (ROS 2 + Gazebo Harmonic).

Related works

Four key papers:

- Kebria et al.: full UR5 kinematic and dynamic model, but fixed payload.
- Schober et al.: vision-based volume estimation, RMSE 21–26 mL, no feedback during the pour.
- Luo et al.: wrist force/torque sensing of batter, open-loop pour.
- Babaians et al. (PourNet): reinforcement-learning pouring with weight feedback, needs heavy training and doesn't model arm dynamics.

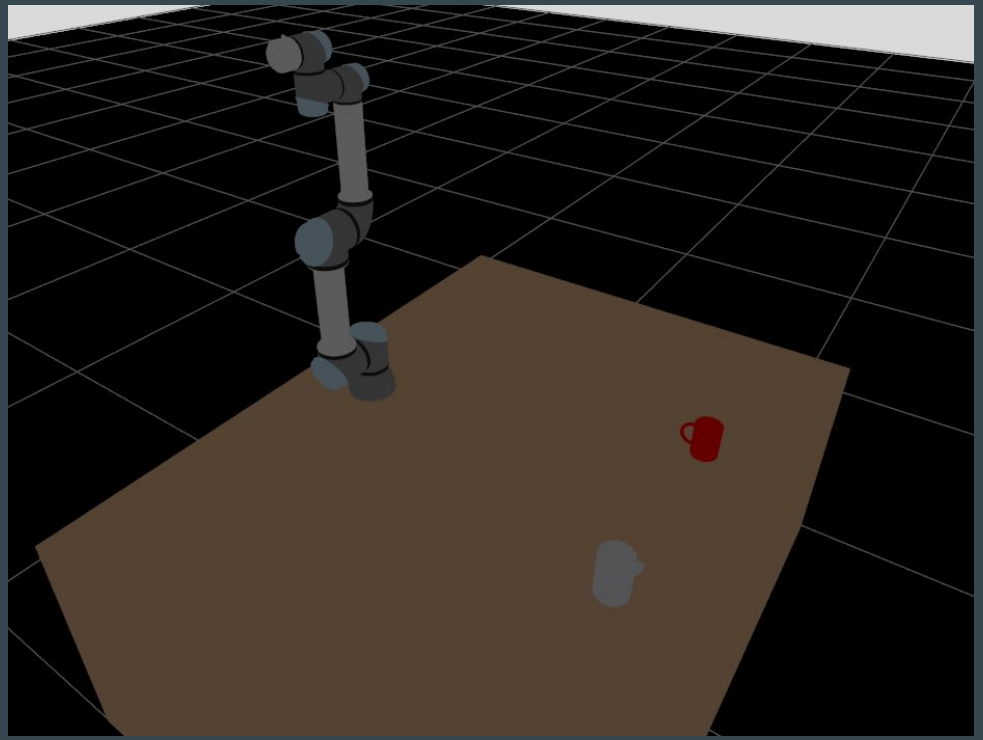
Gap and our approach

TABLE I
COMPARISON OF RELATED WORK AND THIS PROJECT

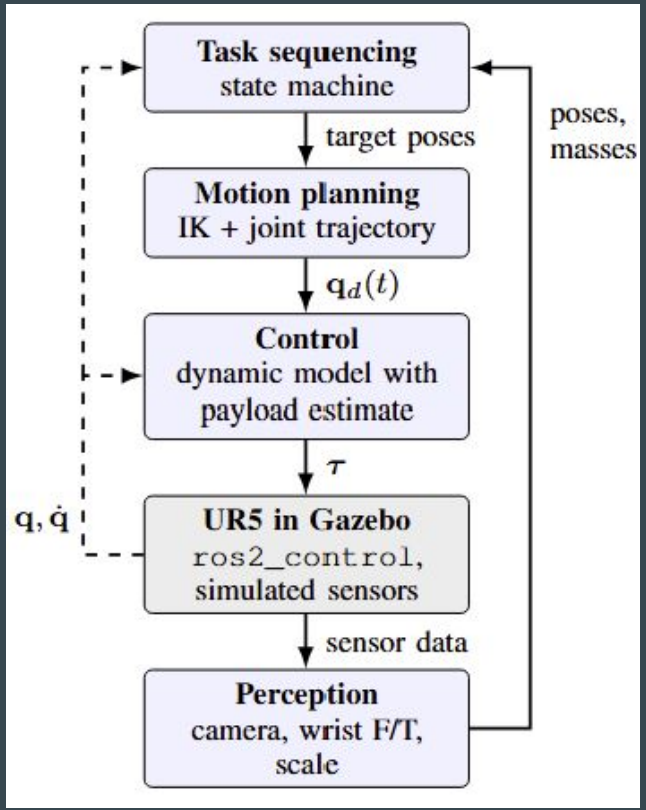
Work	Sensing	Pour control	Time-varying load modelled
[1]	–	PD tracking	No
[2]	RGB camera	Simulation-selected, no feedback	No
[3]	Wrist F/T	Open-loop	No
[4]	F/T (weight of target)	RL + NMPC	Learned implicitly
Ours	Camera, scale, wrist F/T	Model-based, dynamic load	Yes

Test sequence and simulation setup

1. Mug pick-up
2. Mug placement
3. Pitcher pick-up
4. Pour
5. Pitcher return
6. Serving



System pipeline



Kinematic model

TABLE II
UR5 DH PARAMETERS (UR5_PARAMS.PY)

i	a_i [m]	α_i	d_i [m]	θ_i
1	0	$\pi/2$	0.08916	θ_1
2	-0.425	0	0	θ_2
3	-0.39225	0	0	θ_3
4	0	$\pi/2$	0.10915	θ_4
5	0	$-\pi/2$	0.09465	θ_5
6	0	0	0.0823	θ_6

Home configuration

$$\mathbf{q}_{\text{home}} = [0, -\tfrac{\pi}{2}, 0, -\tfrac{\pi}{2}, 0, 0]^T$$

Jacobian

$$\begin{bmatrix} \mathbf{v} \\ \boldsymbol{\omega} \end{bmatrix} = J(\mathbf{q})\dot{\mathbf{q}}$$

$$J_i = \begin{bmatrix} \mathbf{z}_{i-1} \times (\mathbf{p}_e - \mathbf{p}_{i-1}) \\ \mathbf{z}_{i-1} \end{bmatrix}$$

Manipulability (3 singularities)

$$w(\mathbf{q}) = \left| a_2 a_3 \, s_3 \, s_5 \, (a_2 c_2 + a_3 c_{23} + d_5 s_{234}) \right| \quad [\text{m}^3].$$

FK, maps joint angles to tool pose

$$T(\mathbf{q}) = e^{[\mathcal{S}_1]\theta_1} \cdots e^{[\mathcal{S}_6]\theta_6} M$$

IK closed-form solution (up to 8)

$$\theta_1 = \operatorname{atan2}(p_{5y}, p_{5x}) \pm \cos^{-1} \frac{d_4}{\sqrt{p_{5x}^2 + p_{5y}^2}} + \frac{\pi}{2}, \quad (1)$$

$$\theta_5 = \pm \cos^{-1} \frac{p_x s_1 - p_y c_1 - d_4}{d_6}, \quad (2)$$

$$\theta_6 = \operatorname{atan2}\left(\frac{-\hat{y}_x s_1 + \hat{y}_y c_1}{s_5}, \frac{\hat{x}_x s_1 - \hat{x}_y c_1}{s_5} \right). \quad (3)$$

Dynamic model with time-varying payload

Dynamics model

$$M(\mathbf{q}, \boldsymbol{\pi}_p)\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}}, \boldsymbol{\pi}_p)\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}, \boldsymbol{\pi}_p) = \boldsymbol{\tau}$$

Mass, first moment, inertia of the grasped object

$$\boldsymbol{\pi}_p = (m_p,\ \mathbf{h}_p = m_p \mathbf{c}_p,\ I_p)$$

Payload torque enters additively

$$\boldsymbol{\tau} = \boldsymbol{\tau}_{\text{arm}}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}) + J(\mathbf{q})^T \mathcal{W}_p$$

Draining pitcher and payload estimation

- Pitcher = rigid shell + cylinder of liquid. As it drains, mass & center of mass drop
- Mass estimate from the wrist sensor, filtered at 1 Hz

$$\hat{m}_p = \frac{\mathbf{f}^T \mathbf{e}}{\|\mathbf{e}\|^2}, \qquad \hat{m}_l = \hat{m}_p - m_s$$

- Scale gives an independent poured mas, stops the pour and cross-checks m̂_p.
- Draining 0.8 kg changes shoulder/elbow gravity torque by 17% (22.6 → 18.8 N·m). Fixed-payload model carries this error
- Verified by unit tests: M positive definite, Ṁ − 2C skew-symmetric, payload torque = J^T W_p, estimate converges under noise
- Limitation: slosh neglected

Controller

- Cartesian position controller, Jacobian-based.
- Desired tool velocity

$$\mathbf{v}_d = K_p(\mathbf{p}_d - \mathbf{p})$$

- Joint velocities by damped least squares:

$$\dot{\mathbf{q}} = J_v^T(J_v J_v^T + \lambda^2 I)^{-1}\mathbf{v}_d$$

- λ keeps the commands bounded near singularities.
- Commands go to Gazebo through ros2_control.

Status and next steps

- Done: literature review, kinematics, dynamic model with draining payload, payload estimator, pipeline, controller
- Next: full pick-and-pour run in Gazebo, comparing fixed-load vs dynamic-load control (pour error, tracking error, manipulability along the trajectory)